

# User Manual — Tracker Porti

 Manuale in italiano: [manuale.md](#) · [index.html](#)

Complete guide to using **Tracker Porti** as an end user. Explains what you can do and how to do it, screen by screen.

This manual covers day-to-day use of the application. Administration functions (user management, roles, system logs) are not covered here.

## What is Tracker Porti

**Tracker Porti** monitors AIS ship traffic in real time over one or more geographic areas of your choice. It collects the data ships broadcast via AIS, analyzes it, computes a **risk score** for every vessel, and presents it all on interactive maps and tables.

You can:

- **monitor several ports/areas** at once, each with its own data stream;
- **see current ships, past ships**, and **traffic statistics**;
- **follow individual ships** wherever they go, even outside monitored areas;
- consult the **full detail** of every ship (identity, route, sanctions, suspicious behavior...);
- **receive notifications** in-app and on Telegram for relevant events;
- **export** data (CSV, GeoJSON, KML) and take **backups**.

No technical knowledge is required to use it.

## Quick start

1. Open your browser at the server address (e.g. `http://localhost:3000` ).
2. **Log in** with your credentials (see [Account and access](#)).

3. In the left sidebar, choose the **area** to display from the **Area:** dropdown.
4. Press ► **Start monitoring** to start receiving data for that area.
5. Ships appear automatically on the map and in the table.

## Account and access

The application is protected by **login**: you must authenticate to use it. Each user has their own data (areas, followed ships, notifications, settings), separate from other users'.

# Tracker Porti

Accedi al tuo account

Email o username

Password

Accedi

[Registrati](#)

[Password dimenticata?](#)

*Login screen: Email or username and Password fields, with Register and Forgot password links.*

## Registering

From the login page, follow the **Register** link and enter **first name, last name, email, and password**. The new account is created in a *“pending”* state: you’ll be able to log in **only after an administrator approves it**. Once approved, log in normally.

## Logging in

On the login page, enter your password and, as identifier, **either your username or your email** (both work). The session stays valid for several days: normally you don't need to re-enter your credentials on every visit.

After too many failed attempts in a short time (10 in 15 minutes), login is **temporarily blocked** for a few minutes, as anti-bot protection. Wait and try again.

## Logging out

Use the account widget **top right** and choose **Log out**.

## Forgot password

The login page has a **Forgot password?** link. Email delivery is not currently active: to reset your password, **contact an administrator**, who will generate a **one-time link** (valid 24 hours) for you to set a new one.

## Your data

Each user has their **own**:

- monitoring **areas**;
- **settings** (notification preferences, map options, language, default area);
- **flagged ships** ★ and **followed ships**;
- **notifications**.

You see the AIS data of ships located **inside your areas**. Enrichment sources (VesselFinder, MarineTraffic, sanctions, etc.) and the risk-score configuration are instead **shared** and managed by administrators.

If an administrator puts you in a **user group**, some things (areas, followed/flagged/muted ships, notification and map preferences) are **shared with the other members**: what you add shows up for them and vice versa. Your **Telegram link** and interface **language** stay personal.





## First steps (tutorial)

**Just logged in for the first time? Start here.** This quick guide walks you from the first things to do — defining an area, starting monitoring, searching for and following a ship — to the rest of the app. Every step links to the detailed section further in the manual. Once done, you're operational; the rest of the manual is reference material.

### Step 1 — Define your first area

An **area** is the geographic box you want to watch (a port, a stretch of sea). On first login you may not have one yet.

1. In the sidebar, press  **Areas**.
2. Give the area a **name** (e.g. "Port of Bari").
3. Define the boundaries the easy way: pan and zoom the map until it frames the zone, then press  **Capture current view** (or enter the SW and NE coordinates by hand).
4. Press **Add area**: the area is saved and immediately starts receiving data.

## AGGIUNGI AREA

Inserisci le coordinate in gradi decimali, oppure inquadra l'area sulla mappa e premi "Cattura vista corrente".

Nome area

Es. Porto di Ancona

Parola chiave (facoltativa)

Es. ANCONA

SW Lat

40.95

SW Lon

16.60

NE Lat

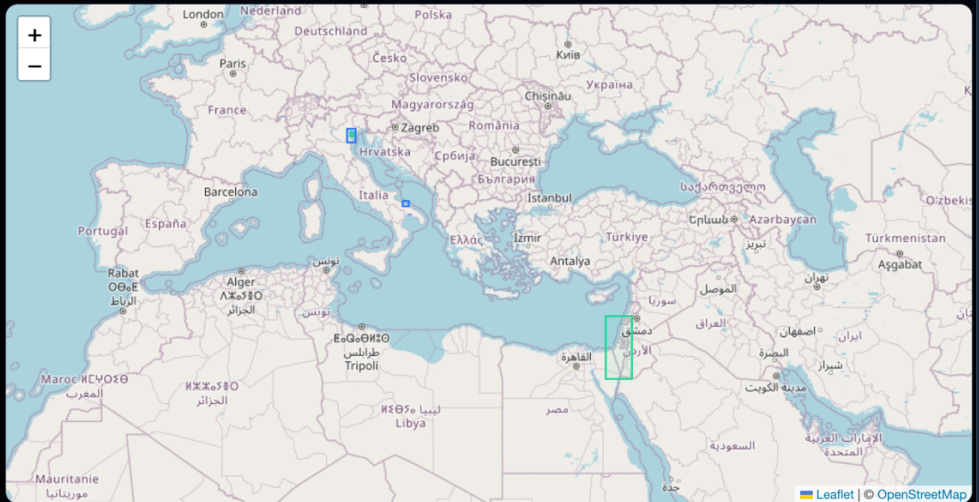
41.30

NE Lon

17.10

Cattura vista corrente

+ Aggiungi area



| NOME                      | SW (LAT, LON)    | NE (LAT, LON)    | PAROLA CHIAVE | STATO  | DATI SALVATI             | AZIONI |
|---------------------------|------------------|------------------|---------------|--------|--------------------------|--------|
| Porto di Bari             | 40.9500, 16.6000 | 41.3000, 17.1000 | BARI          | spenta | 12 letture · 0 navi      |        |
| Israele                   | 29.1801, 33.6732 | 33.6504, 35.8155 | Israele       | attiva | 28.2k letture · 786 navi |        |
| Lido di venezia           | 45.2923, 12.2436 | 45.4938, 12.5045 | Lido          | attiva | 231 letture · 203 navi   |        |
| Nord Adriatico            | 44.9200, 11.9200 | 45.7300, 12.6800 | —             | spenta | 9 letture · 1 navi       |        |
| Porto di Taranto (in uso) | 40.4200, 17.1200 | 40.5200, 17.3000 | TARANTO       | attiva | 5.4k letture · 113 navi  |        |

Areas screen: Add area panel, map with area rectangles, and table with coordinates and status.

→ Full details in [Managing areas](#).

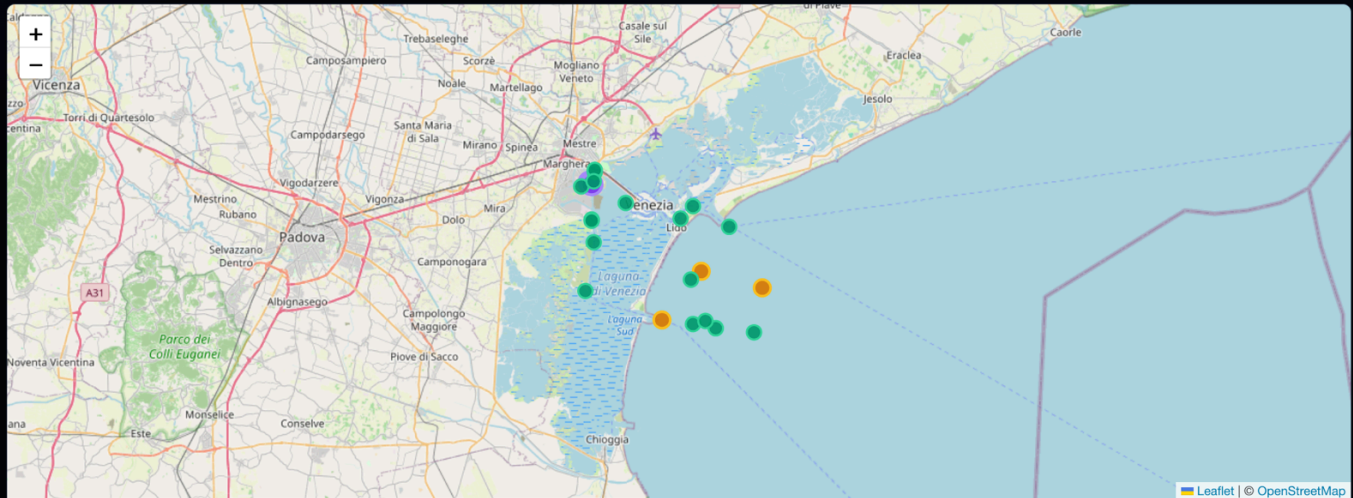
## Step 2 — Start monitoring and watch the ships

1. Go back to monitoring with **Monitoring**.
2. Select the area in the **Area:** dropdown at the bottom of the sidebar.
3. Press ► **Start monitoring**: the ● **ACTIVE** badge appears at the top.
4. Within a short time, ships appear on the **map** (colored by risk) and in the **table** below. Click a row to open the ship's **detail**.

Navi presenti 22

Navi passate 180

Traffico



Cerca nome, MMSI, IMO, destinazi

Tutte le fasce

☐ Solo in porto☐ Solo segnalate☐ Navi segnate come viste☐ Banchine

Esporta...

| FLAG     | ULTIMO AVV.       | NOME NAVE                                                | MMSI      | TIPO NAVE          | DESTINAZIONE      | SOG     | DIREZIONE    | RISCHIO (0-100) |
|----------|-------------------|----------------------------------------------------------|-----------|--------------------|-------------------|---------|--------------|-----------------|
| ★ ✓ 🇮🇹 📄 | 11:18:55 16/07/26 | MARLA BULL <input type="button" value="In porto"/>       | 538010265 | Cargo - Hazmat A ⓘ | Venezia           | 0.0 kn  | ⚓ Ferma      | 66              |
| ★ ✓ 🇮🇹 📄 | 04:24:07 15/07/26 | MSC ELEONORA III <input type="button" value="In porto"/> | 356167000 | Cargo - Hazmat A ⓘ | VENICE            | 0.0 kn  | ⚓ Ferma      | 83              |
| ☆ ✓ 🇮🇹 📄 | 12:16:01 16/07/26 | SAN NICOLO                                               | 247290100 | Passeggeri         | TRONCHETTO<>LI... | 5.9 kn  | ⚓ In entrata | 15              |
| ☆ ✓ 🇮🇹 📄 | 12:12:56 16/07/26 | LADY ESTEY                                               | 319325700 | Diporto            | Venezia           | 10.2 kn | ⚓ In uscita  | 0               |

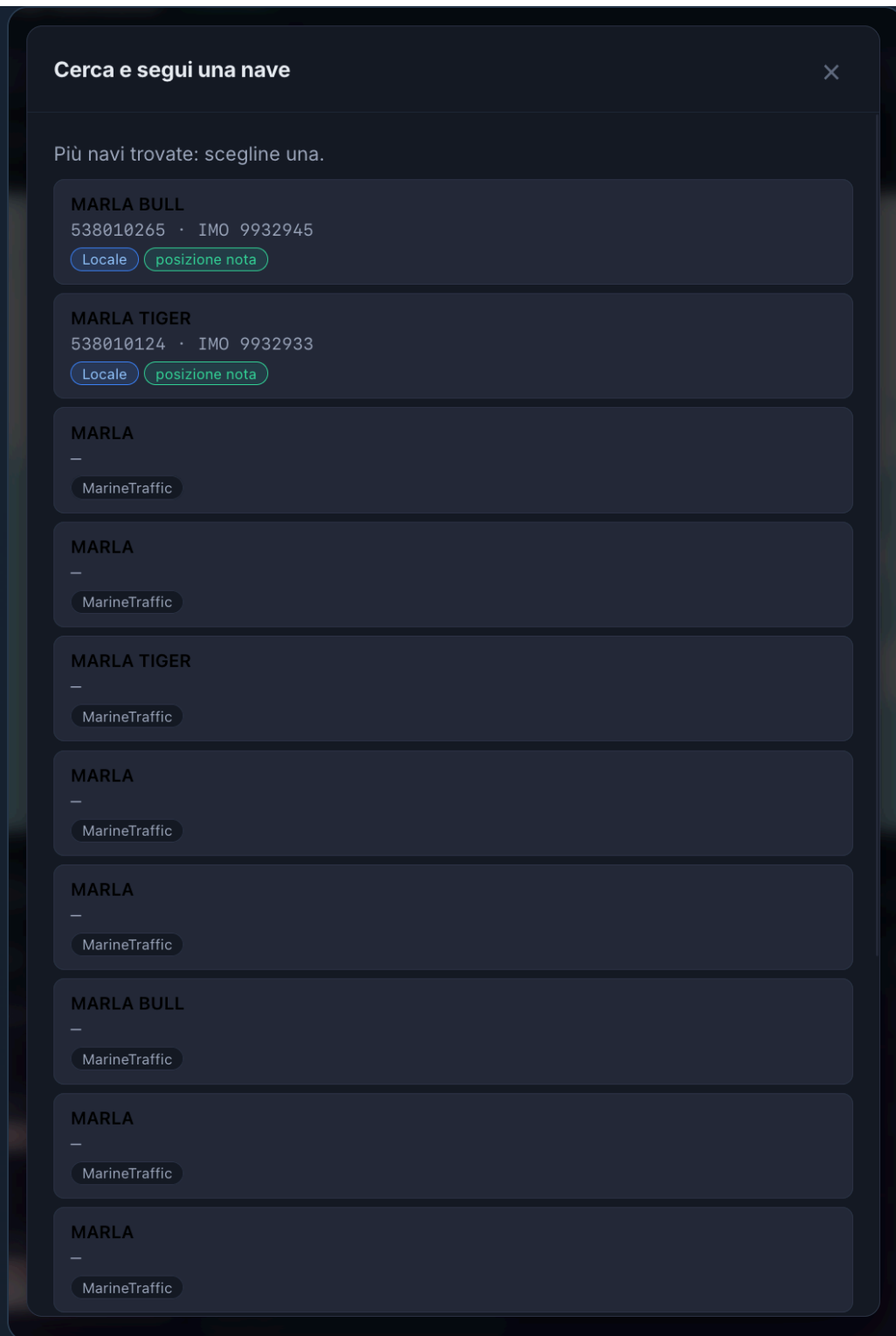
Current ships tab: map with ship markers and detailed table below.

→ Details in [The three main tabs](#) and [Ship detail](#).

### Step 3 — Search for and follow a ship

**Followed ships** are ships you keep an eye on wherever they go, even outside your areas.

1. Sidebar → 📄 **Followed ships**.
2. In the search bar, type the ship's **name** or **MMSI** and press 🔍 **Search**.
3. Wait for the results window to fetch identity and **live position** (up to ~90s).
4. When 📄 **Follow ship** becomes enabled, press it: the ship is added to your followed list.



*Ship search results window: identity, data from external sources, and incoming live position.*

→ Details in [Followed ships](#).

#### Step 4 — Customize (optional)

- ★ **Flag** a ship to highlight it; ✓ **Seen** to mark ones you've already checked.

- Turn on  **Notifications** (ship return, high score, etc.) and, if you want, link **Telegram**.
- Switch light/dark **theme** with the  /  button, bottom right.

→ See [Notifications](#) and [Settings](#).

**That's all it takes to get started.** From here the manual goes into the detail of every feature.

## Main interface

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Notifiche 28

ANNA arrivata in Israele — score rischio 80/100

ANNA arrivata in Israele — score rischio 80/100

ANNA arrivata in Israele — score rischio 80/100

SERAFINA arrivata in Israele — score rischio

Area:

Lido di venezia

Log attività

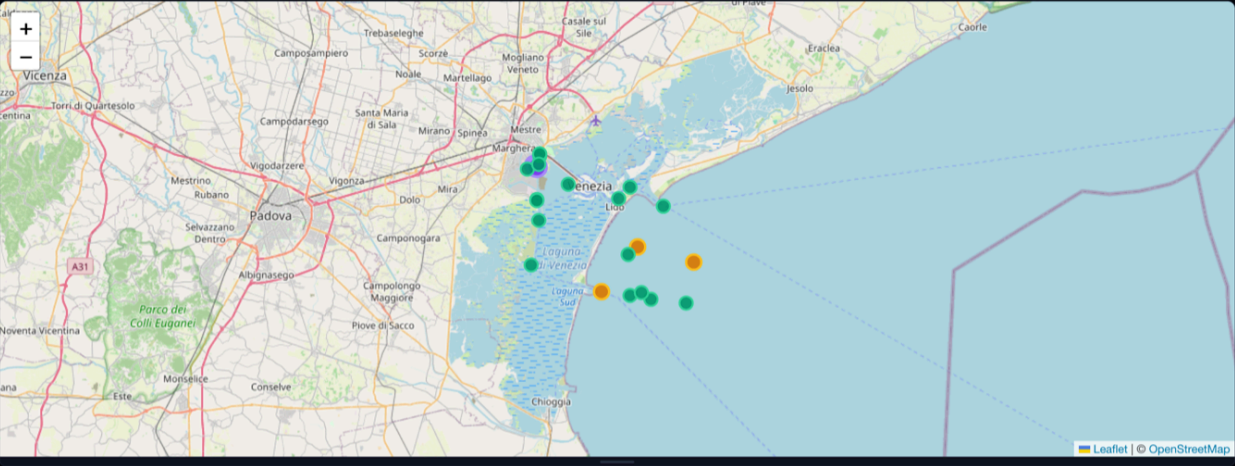
33.807 letture

Tracking porti - Lido di venezia ATTIVO

Navi presenti 22

Navi passate 180

Traffico



Cerca nome, MMSI, IMO, destinaz

Tutte le fasce

Solo in porto

Solo segnalate

Navi segnate come viste

Banchine

Banchine

Banchine...

Replay

Esporta...

Cancella dati

| FLAG                                                                                    | ULTIMO AVV.       | NOME NAVE                              | MMSI      | TIPO NAVE        | DESTINAZIONE      | SOG     | DIREZIONE               | RISCHIO (0-100) |
|-----------------------------------------------------------------------------------------|-------------------|----------------------------------------|-----------|------------------|-------------------|---------|-------------------------|-----------------|
| ★ ✓  | 11:18:55 16/07/26 | MARLA BULL <span>In porto</span>       | 538010265 | Cargo - Hazmat A | Venezia           | 0.0 kn  | <span>Ferma</span>      | 66              |
| ★ ✓  | 04:24:07 15/07/26 | MSC ELEONORA III <span>In porto</span> | 356167000 | Cargo - Hazmat A | VENICE            | 0.0 kn  | <span>Ferma</span>      | 83              |
| ☆ ✓  | 12:16:01 16/07/26 | SAN NICOLO                             | 247290100 | Passeggeri       | TRONCHETTO<>LI... | 5.9 kn  | <span>In entrata</span> | 15              |
| ☆ ✓  | 12:12:56 16/07/26 | LADY ESTEY                             | 319325700 | Diporto          | Venezia           | 10.2 kn | <span>In uscita</span>  | 0               |
| ☆ ✓  | 12:04:59 16/07/26 | HISTRIA GIADA <span>In porto</span>    | 256625000 | Tanker           | Venezia           | 0.0 kn  | <span>Ferma</span>      | 12              |
| ☆ ✓  | 11:55:03 16/07/26 | NEPTUNE ITHAKI <span>In porto</span>   | 215587000 | Cargo            | Venezia           | 0.0 kn  | <span>Ferma</span>      | 26              |
| ☆ ✓  | 11:25:23 16/07/26 | YANG A <span>In porto</span>           | 255003300 | Cargo            | Venezia           | 0.0 kn  | <span>Ferma</span>      | 22              |

Home page: sidebar on the left, map with ships in the center, ships table at the bottom, and notification list.

## Sidebar (left)

| Element                                                                                                   | Function                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Monitoring</b>       | Return to the home (Current / Past ships / Traffic tabs)                                                                                                                                                                                                                                |
|  <b>Start monitoring</b> | Starts real-time AIS data reception for the displayed area                                                                                                                                                                                                                              |
|  <b>Stop</b>             | Stops reception for the current area (already-collected data stays)                                                                                                                                                                                                                     |
|  <b>Clear data</b>       | Deletes the readings for the displayed area — <b>irreversible</b>                                                                                                                                                                                                                       |
|  <b>Areas</b>            | Area management: list, map, add and remove (see <a href="#">Managing areas</a> )                                                                                                                                                                                                        |
|  <b>Coverage map</b>     | World map of AIS coverage (see <a href="#">Coverage map</a> )                                                                                                                                                                                                                           |
|  <b>Settings</b>       | App settings, including technical tabs                                                                                                                                                                                                                                                  |
|  <b>Notifications</b>  | Shows/hides the notification list; a red badge counts unread ones                                                                                                                                                                                                                       |
| <b>Area:</b>                                                                                              | Selects the zone to display. Doesn't start or stop the stream — each area has its own.  = stream active,  = off |
| <b>N readings</b>                                                                                         | Counter at the bottom: positions received during the session                                                                                                                                                                                                                            |

## Status bar (top)

The  **ACTIVE** badge (pulsing green dot) indicates the displayed area's stream is running. **INACTIVE** means reception for that area is stopped.

## Theme switch

The  /  button bottom right toggles dark (default) and light theme.



# The three main tabs

The home shows three tabs: **Current ships**, **Past ships**, and **Traffic**.

## 1. Current ships

Ships detected in the area over the last few hours, with a real-time map and table.

Tracking porti - Lido di venezia

AmministratoreAdminEsci

Navi presenti22

Navi passate180

Traffico

Cerca nome, MMSI, IMO, destinazi...

Tutte le fasce

☐ Solo in porto

☐ Solo segnalate

☐ Navi segnate come viste

☐ Banchine

Banchine

Banchine...

Replay

Esporta...

Cancella dati

| FLAG     | ULTIMO AVV.       | NOME NAVE                                 | MMSI      | TIPO NAVE          | DESTINAZIONE      | SOG     | DIREZIONE    | RISCHIO (0-100) |
|----------|-------------------|-------------------------------------------|-----------|--------------------|-------------------|---------|--------------|-----------------|
| ★ ✓ 🇮🇹 📄 | 11:18:55 16/07/26 | MARLA BULL <a href="#">In porto</a>       | 538010265 | Cargo - Hazmat A ⓘ | Venezia           | 0.0 kn  | ⚓ Ferma      | 66              |
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| ☆ ✓ 🇮🇹 📄 | 12:16:01 16/07/26 | SAN NICOLO                                | 247290100 | Passeggeri         | TRONCHETTO<>LI... | 5.9 kn  | ↙ In entrata | 15              |
| ☆ ✓ 🇮🇹 📄 | 12:12:56 16/07/26 | LADY ESTEY                                | 319325700 | Diporto            | Venezia           | 10.2 kn | ↗ In uscita  | 0               |

Current ships tab: map with markers colored by risk and detailed table below.

**Map:** ships are markers colored by risk band. You can drag the map's bottom edge to resize it. Two buttons in the top-right corner (hover over them to see what they do): 📄 shows the name next to each ship — with few ships on screen it stays **always visible**, with many it only shows **on hover**; ∞ (off by default) shows each ship's **recent track**, with the same logic: permanent if few ships, on hover over the single ship if many. Your choices are remembered for future visits.

**Table — columns:**

| Column       | Meaning                                                                                     |
|--------------|---------------------------------------------------------------------------------------------|
| Action icons | Flag ★, mark as seen ✓, open on VesselFinder 📄                                              |
| Last seen    | Date and time of the last position received                                                 |
| Ship name    | Name of the vessel                                                                          |
| MMSI         | Unique identification code                                                                  |
| Ship type    | Category (cargo, tanker, passenger...)                                                      |
| Destination  | Declared port. UN/LOCODE codes (e.g. ITTAR ) are resolved to the port name (e.g. "Taranto") |
| SOG          | Speed over ground (knots)                                                                   |
| Direction    | ↙ Inbound / ↗ Outbound / ⚓ Stationary                                                       |
| Risk (0–100) | Score (green/yellow/red)                                                                    |

**Sorting:** click a column header to sort; a second click reverses the order (▲/▼). Sort order persists across automatic refreshes.

**Per-row buttons:**

- ☆ / ★ **Flag** — highlights the ship in purple as “to review”. Click again to remove.
- ✓ **Seen** — marks the ship as already reviewed (the row is dimmed). Useful for not missing new arrivals.
- 📄 **VesselFinder** — opens the ship’s page on the VesselFinder site.

**Row colors:**

| Color           | Meaning                      |
|-----------------|------------------------------|
| Red 🚢           | Military ship (auto-flagged) |
| Red             | High risk score (71–100)     |
| Purple          | Manually flagged ship        |
| Dimmed          | Ship marked as “seen”        |
| Badge ⚓ In port | Ship currently moored        |



Click a row to open the [full detail](#).

**Filters (bar above the table):** search by name/MMSI/IMO/destination, filter by risk band, **In port only**, **Flagged only**, **Ships marked as seen**, and the **Berths** checkbox for the mooring overlay. On the right:  **Berths**,  **Export...**, and  **Replay** buttons.

Berths (automatic mooring characterization)

Banchine — Lido di venezia

Ricalcola

Unisci (0)

Caratterizzata quando una categoria supera il 60% su almeno 10 attracchi.

| NOME                                         | CARATTERIZZAZIONE  | ATTRACCHI | FORZA CATEGORIA |  |
|----------------------------------------------|--------------------|-----------|-----------------|--|
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 7         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Servizio/Rimorchio | 195       | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 8         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Passeggeri         | 27        | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 3         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Passeggeri         | 19        | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 3         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 3         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 7         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 6         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Altro              | 29        | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Mista              | 16        | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 3         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Mista              | 11        | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 4         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 3         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 8         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 8         | (automatica)    |  |
| <input type="checkbox"/> Banchina senza nome | Non caratterizzata | 7         | (automatica)    |  |

Berths tab: colored polygons on the map representing moorings characterized by ship category.

Turning on the **Berths** overlay shows where ships dock and what type they are, learned automatically.

How it works:

1. **Mooring detection** — every time a ship stops in port, the system records a mooring point (centroid of the stationary ship's positions in the area).
2. **Grouping** — nearby points are grouped into **berths** (the outline is the envelope of the group's points).
3. **Characterization** — for each berth, ship categories are counted. When one category exceeds **60%** of moorings (out of at least **10** moorings), the berth is **colored** with that category; below the threshold it's **"mixed"** (grey). With fewer than 10 moorings it stays dashed.

#### Using the overlay:

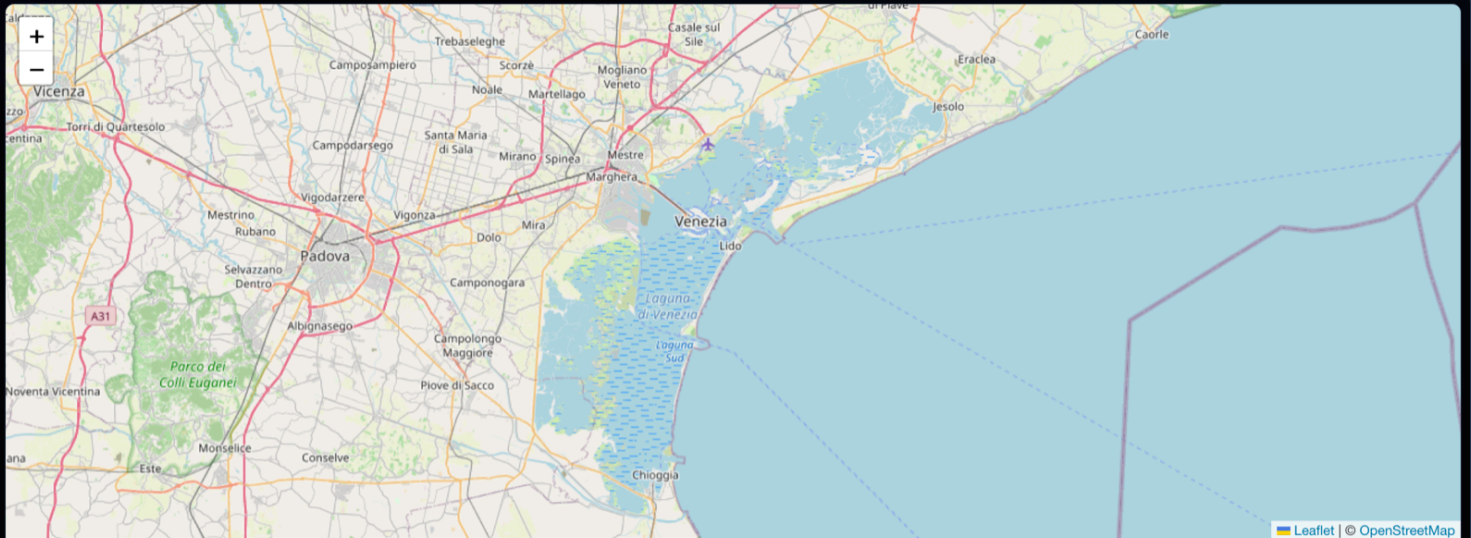
- Check **Berths** in the filter bar to show/hide it (the choice is remembered).
- Every berth has a colored polygon and an always-visible **center dot**.
- **Click a berth** to see its name, characterization, number of moorings, breakdown by category, breakdown by cargo type, and any share of dangerous goods (☼).

**Manual correction** (⚓ **Berths** button): opens the management panel, where you can **rename**, **force the category**, **merge** several berths, **delete** a berth, or **recompute** right away. Clicking a list row centers the map on that berth.

Berths edited by hand (geometry, name, forced category) **survive automatic recomputation**: manual corrections are never overwritten.

**OpenSeaMap overlay.** In settings (→ General, on by default) there are two independent switches: the **Nautical layer (tile)** draws OpenSeaMap nautical symbols (buoys, lighthouses, lights, signals, anchorages) as a single image; **Markers (selectable)** draws ⚓ markers for moorings/berths/official ports from OpenStreetMap, filterable by category. The data is free (no API key) and coverage in commercial ports may be incomplete.

## Historical replay (reviewing past traffic)



*Replay mode: control bar with play/pause, time scrubber, and speed selector, ships moving with trails.*

The ► **Replay** button replays how traffic in the area moved during a past interval. In replay mode, live markers are hidden and a control bar appears:

- **Area** — which of your areas to replay.
- **Window** — presets **1h / 6h / 24h / all**, or a **custom** interval. It automatically snaps to the most recent available data.
- ► / ||, the **scrubber**, and **speeds** (1x / 5x / 20x / 60x).

Each ship moves interpolated between its real positions, leaves a **fading trail**, and is colored by risk; click it for detail. During a **signal gap**, the ship is **hidden** instead of “teleporting”. Press **✕ Exit** to return to live.

If you have the ShipFinder/MyShipTracking integrations enabled, an **Include SF/MST** toggle appears (on by default), which also uses those positions to fill in stretches where AIS is silent.

## 2. Past ships

Tracking porti - Lido di venezia

AmministratoreAdminEsci

Navi presenti22

Navi passate180

Traffico

Cerca nome, MMSI, IMO, destinazione...

Tutte le fasce

☐ Solo segnalate

☐ Navi segnate come viste

Esporta...

Cancelldata

| FLAG     | NOME NAVE         | MMSI      | ULTIMO AVV.       | PRIMO AVV.        | DURATA SOSTA | SOG     | RISCHIO (0-100) |
|----------|-------------------|-----------|-------------------|-------------------|--------------|---------|-----------------|
| ★ ✓ 🇮🇹 📄 | FLORA             | 341561002 | 15:56:16 07/07/26 | 20:17:43 30/06/26 | 163h 39m     | 0.0 kn  | 63              |
| ☆ ✓ 🇮🇹 📄 | WIND SURF         | 309242000 | 19:27:12 15/07/26 | 11:56:10 11/07/26 | 103h 31m     | 9.0 kn  | 6               |
| ☆ ✓ 🇮🇹 📄 | VITA C            | 247450300 | 19:25:46 15/07/26 | 18:00:13 05/07/26 | 241h 26m     | 6.0 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | ECO ITALIA        | 247384200 | 19:07:37 15/07/26 | 17:02:19 01/07/26 | 338h 5m      | 1.6 kn  | 19              |
| ☆ ✓ 🇮🇹 📄 | GIOVANNI CALDERAN | 247370200 | 19:03:50 15/07/26 | 15:58:22 30/06/26 | 363h 5m      | 4.1 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | LEFKA ORI         | 239575000 | 13:07:50 15/07/26 | 15:31:01 08/07/26 | 165h 37m     | 5.8 kn  | 21              |
| ☆ ✓ 🇮🇹 📄 | MORAGA            | 538009489 | 08:06:32 15/07/26 | 20:08:47 04/07/26 | 251h 58m     | 2.3 kn  | 48              |
| ☆ ✓ 🇮🇹 📄 | AMMIANA           | 247290700 | 06:48:02 15/07/26 | 15:47:01 29/06/26 | 375h 1m      | 4.2 kn  | 21              |
| ☆ ✓ 🇮🇹 📄 | METAMAUCO         | 247290200 | 05:09:26 15/07/26 | 15:47:08 29/06/26 | 373h 22m     | 8.2 kn  | 15              |
| ☆ ✓ 🇮🇹 📄 | VANNA C.          | 247191300 | 04:36:46 15/07/26 | 15:51:27 30/06/26 | 348h 45m     | 4.9 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | NYSTED MAERSK     | 477642700 | 02:40:35 15/07/26 | 09:41:35 24/06/26 | 496h 59m     | 11.7 kn | 40              |
| ☆ ✓ 🇮🇹 📄 | KRASZEWSKI        | 636021722 | 20:48:14 14/07/26 | 13:23:07 12/07/26 | 55h 25m      | 10.8 kn | 42              |
| ☆ ✓ 🇮🇹 📄 | AQUILEIA          | 247290600 | 20:47:09 14/07/26 | 17:03:33 04/07/26 | 243h 44m     | 9.5 kn  | 6               |
| ☆ ✓ 🇮🇹 📄 | CARLA BARUZZI     | 247371100 | 20:44:45 14/07/26 | 14:37:19 30/06/26 | 342h 7m      | 0.4 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | MOSE I            | 247354500 | 20:43:38 14/07/26 | 15:44:16 30/06/26 | 340h 59m     | 0.0 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | CANNAREGIO        | 247489600 | 20:31:59 14/07/26 | 15:59:18 30/06/26 | 340h 33m     | 11.0 kn | 0               |
| ☆ ✓ 🇮🇹 📄 | IEVOLI SPEED      | 247351000 | 20:03:44 14/07/26 | 21:24:03 12/07/26 | 46h 40m      | 8.0 kn  | 12              |
| ☆ ✓ 🇮🇹 📄 | DORSODURO         | 247489700 | 19:24:44 14/07/26 | 08:30:11 07/07/26 | 178h 55m     | 0.0 kn  | 0               |
| ☆ ✓ 🇮🇹 📄 | DEO VOLENTE       | 246518000 | 19:20:59 14/07/26 | 20:34:30 13/07/26 | 22h 46m      | 10.7 kn | 15              |
| ☆ ✓ 🇮🇹 📄 | CAN DOLO          | 247300000 | 19:01:17 14/07/26 | 14:00:10 12/07/26 | 20h 30m      | 5.0 kn  | 0               |

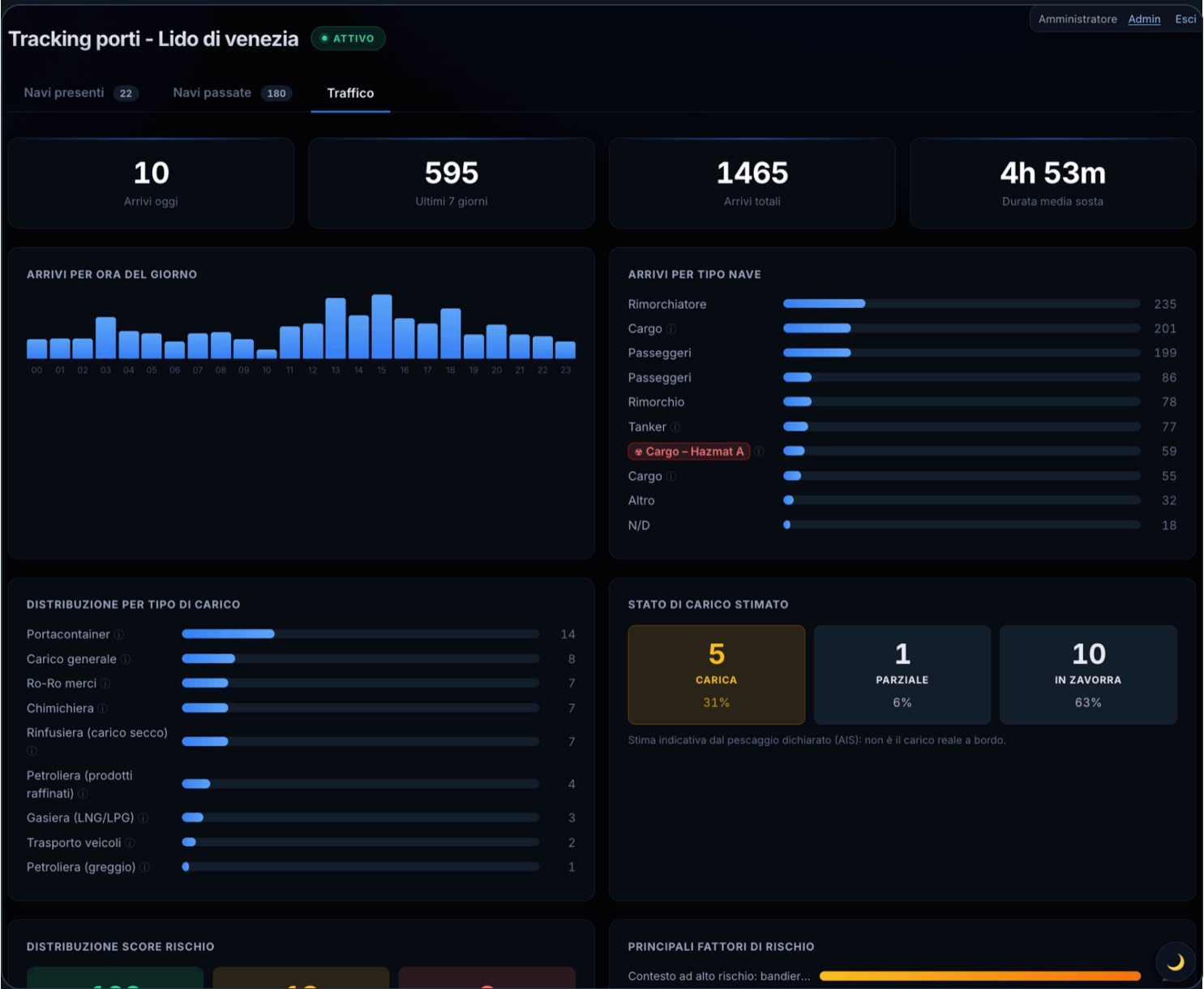
Past ships tab: table of ships that have visited the area, with first contact and stay duration.

Ships that previously visited the area. Table similar to "Current ships", plus:

- **First contact** — when the ship was first seen;
- **Stay duration** — how long it remained in the area.

Column sorting works as in current ships.

3. Traffic



Traffic tab: stat tiles at the top, and charts for arrivals by hour, ship type, risk distribution, and daily trend.

Statistics panel with aggregate indicators and charts.

Tiles at the top: Arrivals today · Arrivals last 7 days · Total arrivals · Average stay duration.

Charts:

| Chart                         | What it shows                |
|-------------------------------|------------------------------|
| Arrivals by hour of day       | Hourly bars (00–23)          |
| Arrivals by ship type         | Ships per category           |
| Risk score distribution       | Breakdown low/medium/high    |
| Top risk factors              | Which factors weigh the most |
| Daily arrivals (last 30 days) | Trend over the month         |
| Highest-scoring ships         | The riskiest vessels         |

**Bottom panels: Expected ships** (heading toward the area, matched by keyword in the destination) and **Latest port events** (recent arrivals/departures).

## Followed ships

The 📶 **Followed ships** section collects ships you follow **wherever they go**, even outside monitored areas, via a dedicated AIS stream. A followed ship that drops off AIS coverage **is not lost**: it stays hooked to a worldwide re-acquisition net and is tracked again as soon as it re-transmits.

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Notifiche 29

ANNA arrivata in israelae — score rischio 80/100

ANNA arrivata in israelae — score rischio 80/100

ANNA arrivata in israelae — score rischio 80/100

SERAFINA arrivata in israelae — score rischio

Area: Lido di venezia

Log attività

Tracking porti - Lido di venezia

Cerca una nave per nome o MMSI da seguire...

Seguite 8Seguite in passato 5

Cerca nome, MMSI, IMO, destinazione...

Tutte le fasce

Solo in porto

Solo segnalate

CSV filtrato

| FLAG | ULTIMO AVV.                                        | NOME NAVE                                                                                                         | MMSI      | TIPO NAVE        | DESTINAZIONE | SOG    | DIREZIONE             | RISCHIO (0-100) |
|------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------|------------------|--------------|--------|-----------------------|-----------------|
| ★ ✓  | 12:27:14 16/07/26                                  | LIFE SUPPORT <a href="#">In porto</a>                                                                             | 352001404 | SAR              | Taranto      | 0.0 kn | <a href="#">Ferma</a> | 20              |
| ★ ✓  | 11:18:55 16/07/26<br>segnale AIS perso da 1 ora    | MARLA BULL <a href="#">In porto</a><br>ShipFinder · 12:21:58 16/07/26<br>MyShipTracking · 12:21:00 16/07/26       | 538010265 | Cargo - Hazmat A | Venezia      | 0.0 kn | <a href="#">Ferma</a> | 66              |
| ★ ✓  | 04:24:07 15/07/26<br>segnale AIS perso da 1 giorno | MSC ELEONORA III <a href="#">In porto</a><br>ShipFinder · 12:26:59 16/07/26<br>MyShipTracking · 10:05:00 16/07/26 | 356167000 | Cargo - Hazmat A | VENICE       | 0.0 kn | <a href="#">Ferma</a> | 83              |
|      |                                                    | ROMAN E <a href="#">In porto</a>                                                                                  |           |                  |              |        |                       |                 |

Followed ships section, Followed sub-tab: list of currently tracked ships with status badges.

Two sub-tabs: **Followed** (currently tracked) and **Previously followed** (history; a ship ends up here only after a very long silence — default ~6 months — or when you stop following it).



Tracking porti - Lido di venezia

ATTIVO

AmministratoreAdminEsci

marla

Cerca

Seguite8Seguite in passato5

Cerca nome, MMSI, IMO, destinazione...Tutte le fasceSolo segnalateCSV filtrato

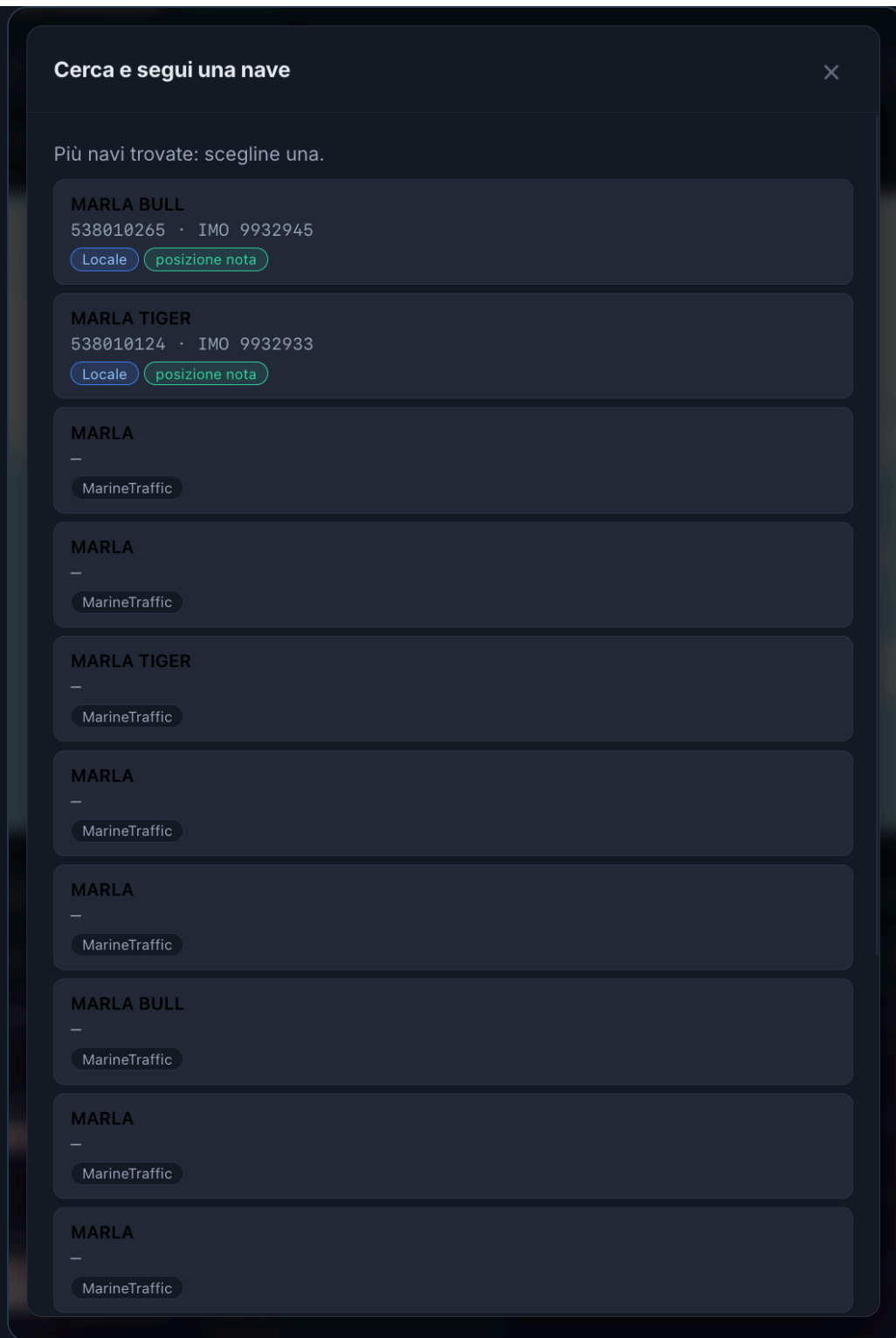
| FLAG    | NOME NAVE        | MMSI      | ULTIMO AVV.       | SOG     | RISCHIO (0-100) |
|---------|------------------|-----------|-------------------|---------|-----------------|
| ★ ✓ 🚢 📍 | FLORA            | 341561002 | 15:56:16 07/07/26 | 0.0 kn  | 63              |
| ☆ ✓ 🚢 📍 | MEDKON MERSIN    | 352001710 | 10:23:47 15/07/26 | 10.9 kn | 72              |
| ☆ ✓ 🚢 📍 | SIRIOS CEMENT II | 356056000 | 18:37:02 14/07/26 | 11.5 kn | 47              |
| ☆ ✓ 🚢 📍 | SKY LIGHT        | 677064500 | 20:03:23 27/06/26 | 8.5 kn  | 100             |
| ☆ ✓ 🚢 📍 | MY WAY           | 247284260 | 09:57:11 16/07/26 | 6.6 kn  | 0               |

Previously followed sub-tab: history of ships no longer tracked.

On the **Followed** sub-tab's map, two buttons in the top-right corner (📄 and 📍, hover over them to see what they do) show/hide the **name next to each marker** and a small **recent-track breadcrumb**, useful to see where a ship is coming from. Your choice is remembered for future visits.

## Searching for and following a ship

At the top of the section there's a **search bar**: type **name** or **MMSI** and press 🔍 **Search**.



*Ship search results window: identity, data from external sources, and incoming live position.*

1. A results window opens that stays open while we gather data. If the name matches several ships, pick the right one.
2. The card fills in **progressively**: identity and data from VesselFinder / MarineTraffic / Global Fishing Watch (with an icon showing where it was found), any **sanctions** or **PSC** alerts, and the **live position** on a mini-map.

3. The position is fetched in real time from AISstream: it can take up to ~90s. If the ship isn't transmitting, a warning with  **Retry** appears.
4. When the position is available,  **Follow ship** becomes enabled: click it to add it to your followed list.

Closing the window (**Cancel**, **✕**, clicking outside, or **Esc**) stops the search without following anything.

## Re-following a ship from “Previously followed”

When you re-follow a ship that was in **Previously followed** (open its detail and press  **Follow ship**), the app **immediately** puts it back among the followed ones and starts searching for its position in the background:

- if the ship is transmitting, it stays among the followed ones and the position updates;
- if it does **not** transmit within ~90 seconds, it **goes back to “Previously followed”** and you get a **notification** that it wasn't found.

## Ship detail

Clicking any table row (or a ship's notification) opens the full detail card, organized into **tabs**: **General**, plus one tab per enabled external source (VesselFinder, MarineTraffic, ShipFinder, MyShipTracking, Equasis, Global Fishing Watch). A source disabled in Settings has no tab.

Tracker

Monitoring

Followed ships

Coverage map

Start monitoring

Stop

Areas

Settings

Notification 85

New berth detected in Italy

01:21:05 25/07/26

New berth detected in Italy

01:21:05 25/07/26

Berth Unnamed berth characterised as Mixed in Italy

01:21:05 25/07/26

Berth Unnamed berth characterised as Mixed in Italy

01:21:05 25/07/26

Area:

Italy — on

Admin manual

User manual

Activity log

3035 readings

Port Tracker - Italy

ACTIVE

Back

G3

MMSI: 319355000

☆

✓

📄

Mark as military vessel

VesselFinder

MarineTraffic

MyShipTracking

ShipFinder

Report

General

VesselFinder

MarineTraffic

ShipFinder

MyShipTracking

Equasis

Global Fishing Watch

RISK SCORE

SHIP TYPE

CARGO TYPE

LOAD STATE

CALL SIGN

IMO

DESTINATION

ETA

MAX DRAUGHT

LENGTH

BEAM

SPEED (SOG)

0

—

—

—

—

—

—

—

—

—

0.0 kn

COURSE (COG)

NAV. STATUS

DIRECTION

IN PORT

POSITION

STAY DURATION

FIRST CONTACT

LAST CONTACT

57.0°

Moored

Stationary

Yes

43.69750°N, 7.28433°E

39m

00:42:04 25/07/26

01:21:05 25/07/26

RISK FACTORS (SCORE 0/100)

No anomalies detected - regular navigation

+

-

Parc naturel régional des Préalpes d'Azur

Carros

Vence

Grasse

Valbonne

Mouans-Sartoux

Mougins

Le Cannet

Antibes

La Trinité

Monaco

Menton

Ventimiglia

Camporosso

Sanremo

Imperia

Diano Marina

Leaflet

OpenStreetMap

Export track:

Esporta...

Reset replay

AGGREGATED SHIP DATA (ALL PROVIDERS)

Ship name

G3

IMO

9508237

MMSI

319355000

Call sign

ZCTR

Flag

CYM

Ship detail — header, tabs, information grid, map, and aggregated ship data table (General tab).

## Header and actions

- ← **Back** — return to the list
- ★ **Flag** / ✓ **Seen** — flag / mark as reviewed
- 🛡️ **Mark as military ship** — classifies the ship as military (red row, maximum risk)
- 🔔 / 🔕 — mute or re-enable automatic notifications for this ship
- 🔗 **VesselFinder** / **MarineTraffic** / **ShipFinder** / **MyShipTracking** — open the external page
- Report** — generates a report for the ship

## General tab

The first tab, open by default, gathers everything that **isn't specific to a single external source**: information grid, risk factors, map with track and replay, aggregated ship data table (below), AIS readings, operational notes, visit history in monitored areas, and — when present — sanctions and rendezvous at sea. Each enabled external source instead has its own dedicated tab (see below).

# Information grid

All available data for the ship:

| Field                | Meaning                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| Risk score           | Colored badge 0–100                                                                                            |
| Ship type            | Category; "☢ Hazmat" if carrying dangerous goods                                                               |
| Cargo type           | Cargo class (container ship, tanker, chemical tanker, gas carrier, bulk carrier...) with source in parentheses |
| Load status          | Estimated laden / partial / ballast from declared draught                                                      |
| Call sign            | Radio call sign                                                                                                |
| IMO                  | IMO registration number                                                                                        |
| Destination          | Declared port                                                                                                  |
| ETA                  | Estimated time of arrival                                                                                      |
| Max draught          | Hull depth in water (m)                                                                                        |
| Length / Beam        | Physical dimensions                                                                                            |
| SOG / Course         | Current speed and heading                                                                                      |
| Nav status           | Moored, underway, etc.                                                                                         |
| Direction            | Inbound / outbound / stationary                                                                                |
| Position             | Last latitude and longitude                                                                                    |
| Stay duration        | Time elapsed since arrival                                                                                     |
| First / Last contact | Timestamp of the first and last data received                                                                  |

# Risk factors

DATI MARINETRAFFIC

cache · 12:39:13 16/07/26

Nome

MARLA BULL

IMO

9932945

MMSI

538010265

Nominativo

V7A5823

Bandiera

MARSHALL ISLANDS

Tipo

Container Ship

Stazza lorda

18658

Portata lorda (DWT)

22396

Lunghezza (m)

171.98

Larghezza (m)

28.46

Anno costruzione

2022

Stato

Active

Porto di armamento

MAJURO

DATI SHIPFINDER

cache · 12:42:47 16/07/26

Localizza via ShipFinder

Ultima posizione nota

45.36868, 12.75537

SOG 9.8 kn · COG 173° · 12:42:38 16/07/26

Nome

MARLA BULL

IMO

9932945

MMSI

538010265

Call Sign

V7A5823

Bandiera

MHL

Tipo

Container ship

Lunghezza

171 m

Larghezza

28 m

Pescaggio

7.4 m

Stato

Under way using engine

Ship detail — list of risk factors with the points assigned by each.

List of the factors that contributed to the score, with each one's points. If there are no anomalies, "No anomalies detected" is shown.

## Aggregated ship data (all providers)

Below the map, the General tab shows a table gathering the **main ship data as reported by each enabled external source** — name, IMO, MMSI, call sign, flag, type, year built, length, beam, draught, gross tonnage, deadweight, home port — without having to open every provider's tab to compare them.

- When several sources report **the same value** — even spelled differently (e.g. flag "PAN" and "Panama", length "202.80" and "203") — it appears **once**, with a **colored dot per confirming source** next to the

value.

- When sources **disagree**, **every distinct value** reported appears instead, each with its own colored dot, and the row is highlighted with a light tint so it stands out at a glance: it's a sign one of the sources likely has a wrong or stale value, worth checking.
- **Hover over a dot** to see which source it is — the same color used to tell apart the respective tabs (VesselFinder, MarineTraffic, ShipFinder, MyShipTracking, Equasis, Global Fishing Watch).
- Fields that change often (destination, ETA, live draught, nav status) **don't** appear in this table: they stay in the information grid and in each provider's own tab, where the time of the reading matters.

The table only appears once **at least one source** has data for the ship; it stays empty (hidden) until a source has responded.

## VesselFinder / MarineTraffic tab

If enabled in settings, the respective tab shows additional information fetched from these services (flag, gross tonnage, year built...), noting whether it's cached. Fetching happens automatically in the background for recently seen ships.

# ShipFinder and MyShipTracking tabs (re-locating followed ships)

DATI MYSHIPTRACKINGcache · 12:42:41 16/07/26

Localizza via MyShipTracking

Ultima posizione nota 45.38456, 12.25527 SOG 9.3 kn · COG 185° · 12:41:00 16/07/26

|              |                        |
|--------------|------------------------|
| Nome         | MARLA BULL             |
| IMO          | 9932945                |
| MMSI         | 538010265              |
| Call Sign    | V7A5823                |
| Bandiera     | MH                     |
| Tipo         | Cargo A                |
| Dimensioni   | 171 × 28 m             |
| Pescaggio    | 7.4 m                  |
| Stato        | Under way using engine |
| Destinazione | RAVENNA                |

PROPRIETÀ / GESTIONE (EQUASIS)memorizzato · 11:00:06 30/06/26

PROPRIETÀ E GESTIONE

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Ship manager/Commercial manager | MARLA DRY BULK SHIPMANAGEMENT<br>4, Xenias Street, Kifisia, 145 62, Athens, Greece.<br>since 29/11/2022                             |
| ISM Manager                     | MARLA DRY BULK SHIPMANAGEMENT<br>4, Xenias Street, Kifisia, 145 62, Athens, Greece.<br>since 29/11/2022                             |
| Registered owner                | ADDISON MARINE INC<br>C/O: Marla Dry Bulk Shipmanagement Inc 4, Xenias Street, Kifisia, 145 62, Athens, Greece.<br>since 29/11/2022 |

DATI NAVE

|            |                  |
|------------|------------------|
| Name       | MARLA BULL       |
| IMO number | 9932945          |
| Flag       | Marshall Islands |
| Call Sign  | V7A5823          |
| MMSI       | 538010265        |

Ship detail — additional external source tabs (ShipFinder / MyShipTracking) with last known position.

If you enable **Import ShipFinder** and/or **Import MyShipTracking**, the corresponding tabs appear. Besides static data, these sources provide the **last-sighting position**, used to **re-locate followed ships AIS can no longer see**:

- Automatic** — for every followed ship that hasn't transmitted in a while, the app periodically queries these sources in the background. If it finds a position, it appears on the mini-map (General tab) as a distinct marker (**orange** = ShipFinder, **teal/cyan** = MyShipTracking), without altering the AIS track, score, or replay.
- Dedicated badge** — when a position exists, a  **seen on ShipFinder/MyShipTracking** badge appears next to the name, distinct from the yellow  **searching** badge (which reflects AIS status). The "searching" badge only turns off with a real AIS signal.



- **Manual** — the  **Locate via ShipFinder / MyShipTracking** button fetches the current position **right away**.

The tabs appear **only if the integration is enabled** (Settings → Import...). Off by default. On the Followed ships map, a ship AIS can no longer see is shown at its most recent SF/MST position (grey marker) and returns to live AIS as soon as it re-transmits.

## Equasis tab (ownership / management)

GLOBAL FISHING WATCH cache · 12:39:20 16/07/26

AmministratoreAdminEsci

IDENTITÀ (GFW)

|             |                  |
|-------------|------------------|
| Bandiera ⓘ  | Marshall Islands |
| Call sign ⓘ | V7A5823          |
| IMO ⓘ       | 9932945          |
| MMSI ⓘ      | 538010265        |
| Nome        | MARLA BULL       |

SOSTE IN MARE APERTO (LOITERING) ⓘ

| DATA              | DURATA | POSIZIONE      |
|-------------------|--------|----------------|
| 23:59:31 25/07/25 | 1 h    | 34.161, 25.416 |
| 17:22:07 10/02/26 | 16 h   | 32.272, 33.422 |
| 08:17:24 09/02/26 | 22 h   | 33.329, 31.439 |

SCALI IN PORTO ⓘ

| DATA              | PORTO         |
|-------------------|---------------|
| 21:34:20 27/11/25 | ISR           |
| 15:54:44 23/11/25 | Italy         |
| 06:33:37 21/11/25 | Italy         |
| 04:49:26 19/11/25 | SVN           |
| 21:09:11 14/11/25 | ISR           |
| 11:15:59 13/11/25 | ISR           |
| 00:33:08 09/11/25 | VENICE, Italy |
| 06:21:58 08/11/25 | Italy         |
| 04:19:46 04/11/25 | SVN           |
| 03:34:28 31/10/25 | ISR           |
| 17:55:27 29/10/25 | ISR           |
| 07:24:56 25/10/25 | Italy         |
| 10:14:00 24/10/25 | Italy         |

Ship detail — Equasis tab (Ownership / management), with the fetch button and owner/manager/operator data.

If the Equasis lookup is enabled, the **Equasis** tab appears with a **Fetch Equasis information** button. **Never runs automatically**: the lookup only happens on click and queries Equasis by **IMO number**. It returns ship

data, **ownership and management** (owner, ISM manager, operator), classification, P&I coverage, performance/risk indicators, and recent positions. The result is stored once and shown with no expiry.

## Global Fishing Watch tab

If GFW enrichment is enabled (on by default), the **Global Fishing Watch** tab appears with the ship's **identity** and tables of **behavioral events** derived from the global AIS feed:

- **Encounters** — two ships meeting in open water (a transshipment signature).
- **Loitering** — prolonged stop in open water.
- **Port visit** — reconstructed port calls.
- **AIS off (gap)** — transponder off while underway ("dark activity").

Every event table is **sortable**: click a column header to sort, click again to reverse (default sort is by date, most recent first). Tables with more than 10 rows are **paginated**, with < **Prev** and **Next** > buttons at the bottom.

Enrichment is **proactive** (no button). GFW mainly tracks fishing, support, and reefer/carrier vessels: many merchant ships aren't present (a "not found in GFW" note). These events **feed the risk score**.

# Sanctions

| ORA               | TIPO               | SOG    | COG    | PRUA | LAT      | LON      | STATO NAV.                   |
|-------------------|--------------------|--------|--------|------|----------|----------|------------------------------|
| 12:42:38 16/07/26 | ShipfinderPosition | 9.8 kn | 172.9° | 171° | 45.36060 | 12.25537 | Stato Under way using engine |
| 12:41:00 16/07/26 | ShipfinderPosition | 9.3 kn | 184.5° | —    | 45.36456 | 12.25527 | Stato Under way using engine |
| 12:21:58 16/07/26 | ShipfinderPosition | 7.9 kn | 176.6° | 177° | 45.40603 | 12.26142 | Stato Under way using engine |
| 12:21:00 16/07/26 | ShipfinderPosition | 7.7 kn | 175.6° | —    | 45.40820 | 12.26125 | Stato Under way using engine |
| 11:18:55 16/07/26 | PosReport          | 0.0 kn | 180.6° | 119° | 45.45802 | 12.25555 | Vela                         |
| 11:18:55 16/07/26 | PosReport          | 0.0 kn | 180.6° | 119° | 45.45802 | 12.25555 | Vela                         |
| 10:39:00 16/07/26 | ShipfinderPosition | —      | 119.0° | —    | 45.45801 | 12.25555 | Stato Moored                 |
| 10:36:56 16/07/26 | ShipfinderPosition | 0.0 kn | 180.6° | 119° | 45.45800 | 12.25555 | Stato Moored                 |
| 18:16:31 15/07/26 | StaticData         | —      | —      | —    | 45.45801 | 12.25554 | —                            |
| 18:16:31 15/07/26 | StaticData         | —      | —      | —    | 45.45801 | 12.25554 | —                            |
| 17:58:32 15/07/26 | StaticData         | —      | —      | —    | 45.45806 | 12.25552 | —                            |
| 17:58:32 15/07/26 | StaticData         | —      | —      | —    | 45.45806 | 12.25552 | —                            |
| 16:52:13 15/07/26 | StaticData         | —      | —      | —    | 45.45803 | 12.25557 | —                            |
| 16:52:13 15/07/26 | StaticData         | —      | —      | —    | 45.45803 | 12.25557 | —                            |
| 07:46:12 15/07/26 | StaticData         | —      | —      | —    | 45.45801 | 12.25550 | —                            |
| 07:46:12 15/07/26 | StaticData         | —      | —      | —    | 45.45801 | 12.25550 | —                            |
| 06:57:12 15/07/26 | StaticData         | —      | —      | —    | 45.40184 | 12.26173 | —                            |
| 06:57:12 15/07/26 | StaticData         | —      | —      | —    | 45.40184 | 12.26173 | —                            |
| 06:40:12 15/07/26 | StaticData         | —      | —      | —    | 45.36213 | 12.25482 | —                            |
| 06:40:12 15/07/26 | StaticData         | —      | —      | —    | 45.36213 | 12.25482 | —                            |
| 06:36:54 15/07/26 | StaticData         | —      | —      | —    | 45.35461 | 12.25731 | —                            |
| 06:36:54 15/07/26 | StaticData         | —      | —      | —    | 45.35461 | 12.25731 | —                            |
| 06:31:54 15/07/26 | ShipfinderPosition | 9.8 kn | 284.9° | 286° | 45.34628 | 12.27073 | Stato Under way using engine |

Ship detail — Sanctions panel with red border: list, program, and match field.

When a ship matches a sanctions list, the **Sanctions** panel appears at the top of the detail — with a red border:

- **List** — the matching regime: OFAC SDN (USA), EU consolidated list, UK OFSI, or UN.
- **Programme** — the specific sanctions programme, if available.
- **Matched by** — the field the match was made on: **IMO** or **call sign** (high confidence) or **name** (weaker, possible homonym).
- **Listed name, flag, owner, and aliases** of the entity, when available.

A notice reminds you to always verify against the official source (a **name-only** match can be a false positive). When an identifier is available, **Open official profile** opens the public page. The panel appears **only** for listed ships.

## Rendezvous at sea

If the ship has had a confirmed **rendezvous** with another one (stayed close, slow, and offshore long enough — a ship-to-ship transshipment signature), the **Rendezvous at sea** section appears with the list of encounters (other ship, date/time, minimum distance, area). Every row is clickable and opens the ship involved. A confirmed rendezvous triggers a **notification** and **adds risk points to both** ships.

## Position map

Map with the ship's track and animated playback controls.

- **Time window** — presets **6h / 24h / 7d / all**, or a **custom** interval (From → To, then **Apply**).
- **▶ / ||** — plays/pauses the track animation.
- **Scrubber** — jumps to any point on the track.
- **Speed** — **1× / 5× / 20× / 60×** (default 20×), changeable during playback.
- **Include SF/MST** — if the integrations are active and the ship has scraped positions, includes them in the track (amber nodes = ShipFinder, teal = MyShipTracking).

The **last known position** markers for ShipFinder/MyShipTracking shown on the map follow the **same time window** selected for the track (preset, custom interval, or replay segment): narrowing the window only shows sightings that fall within it, exactly as with AIS positions. The  **Locate via ...** button always shows the just-fetched position regardless.

## AIS readings

### NOTE OPERATIVE

Salva note

#### ANDAMENTO SCORE DI RISCHIO



## STORICO VISITE IN PORTO

AISSTREAM / SHIPEINDER / REPORT POSIZIONE

| EVENTO     | DATA/ORA          | DESTINAZIONE | PESCAGGIO             | DURATA SOSTA |
|------------|-------------------|--------------|-----------------------|--------------|
| 🏠 Arrivo   | 11:18:55 16/07/26 | Venezia      | 6.9 m                 | In porto     |
| 🏠 Arrivo   | 16:00:47 15/07/26 | Venezia      | 6.9 m                 | 2h 39m       |
| 🚪 Partenza | 18:39:45 15/07/26 | Venezia      | 6.9 m 0.0m ↓ scarico  | —            |
| 🏠 Arrivo   | 09:07:33 08/07/26 | ILTHB        | 6.8 m                 | 23h 38m      |
| 🚪 Partenza | 08:45:14 09/07/26 | Koper        | 7.3 m ++0.5m ↑ carico | —            |
| 🏠 Arrivo   | 02:47:55 01/07/26 | Venezia      | 9.1 m                 | 12h 58m      |
| 🚪 Partenza | 15:46:16 01/07/26 | Venezia      | 9.1 m 0.0m ↓ scarico  | —            |
| 🏠 Arrivo   | 22:01:24 29/06/26 | Haifa        | 7.3 m                 | 2h 18m       |
| 🚪 Partenza | 00:19:14 30/06/26 | Haifa        | 7.3 m 0.0m ↓ scarico  | —            |
| 🏠 Arrivo   | 18:48:54 20/06/26 | —            | —                     | 24h 12m      |
| 🚪 Partenza | 19:01:22 21/06/26 | Haifa        | 7.3 m —               | —            |

*Ship detail — paginated table of AIS readings and operational notes section.*

Paginated table with all positions received in chronological order. Click a row for detail. Full raw JSON data is shown only for static messages (name, dimensions, destination...); for simple position messages, the useful fields are already in the grid. Navigate with **← Prev** and **Next →**.

## Operational notes

Free-text area: write any notes about the ship and press **Save notes**. Notes are persisted in the database.

## Visit history in monitored areas

Log of all arrivals (↙) and departures (↗) detected for the ship, with the **monitored area** they occurred in, destination, draught, and stay duration. UN/LOCODE codes in the destination are resolved to the port name (e.g. ITNAP → "Napoli").

The ⓘ icon next to the title explains the concept: a “monitored area” is the geographic box you configured (see [Managing areas](#)), not necessarily a single real port — it can cover an entire gulf with several harbours, or just one berth. An “arrival” is logged when the ship reappears in the area after being absent more than 60 minutes (or is seen there for the first time); a “departure” when it stops being seen while still logged in the area.

## Managing areas

Open with 🗺️ **Areas**. Here you add and remove monitored areas **without restarting the app**.

Tracking porti - Lido di venezia

• ATTIVO

Amministratore Admin Esc

AGGIUNGI AREA

Inserisci le coordinate in gradi decimali, oppure inquadra l'area sulla mappa e premi "Cattura vista corrente".

Nome area

Es. Porto di Ancona

Parola chiave (facoltativa)

Es. ANCONA

SW Lat

40.95

SW Lon

16.60

NE Lat

41.30

NE Lon

17.10

Cattura vista corrente

+ Aggiungi area

| NOME                      | SW (LAT, LON)    | NE (LAT, LON)    | PAROLA CHIAVE | STATO  | DATI SALVATI             | AZIONI |
|---------------------------|------------------|------------------|---------------|--------|--------------------------|--------|
| Porto di Bari             | 40.9500, 16.6000 | 41.3000, 17.1000 | BARI          | spenta | 12 letture · 0 navi      |        |
| Israele                   | 29.1801, 33.6732 | 33.6504, 35.8155 | Israele       | attiva | 28.2k letture · 786 navi |        |
| Lido di venezia           | 45.2923, 12.2436 | 45.4938, 12.5045 | Lido          | attiva | 231 letture · 203 navi   |        |
| Nord Adriatico            | 44.9200, 11.9200 | 45.7300, 12.6800 | —             | spenta | 9 letture · 1 navi       |        |
| Porto di Taranto (in uso) | 40.4200, 17.1200 | 40.5200, 17.3000 | TARANTO       | attiva | 5.4k letture · 113 navi  |        |

Areas screen: Add area panel, map with area rectangles, and table with coordinates and status.

The screen contains:

- an “Add area” panel (top left);
- a map with all areas as rectangles (green = stream active, purple = area in view, blue = others);

- a **table** with name, SW and NE coordinates, keyword, stream status, saved data, and delete button.

## Adding an area

1. Write a **name** (required) and, if you want, a **keyword** (for the “Expected ships” filter).
2. Specify the boundaries in **one** of two ways:
  - **GPS coordinates** — enter by hand the latitude and longitude in **decimal degrees** of the two SW (South-West) and NE (North-East) corners. E.g.: SW 40.95, 16.60 — NE 41.30, 17.10 . A preview rectangle appears as you type.
  - **From map** — frame the area by panning/zooming the map, then press 📍 **Capture current view**: the coordinates fill in on their own.
3. Press **Add area**. The area is saved and its stream starts immediately.

Latitude ranges from -90 to 90 (positive to the North), longitude from -180 to 180 (positive to the East).  
Corners can be entered in any order: they are automatically reordered.

## Removing an area

Press 🗑️ on the area’s row. **Removing the area also deletes all related history** (readings, ships, and port events). For safety, deletion **is not immediate**: a warning appears with a countdown and an ↶ **Undo** button for **10 seconds**. It becomes final when the time expires or when you leave the Areas page.

**At least one area** must remain: the delete button is disabled when only one is left.

# Coverage map

Tracking porti - Lido di venezia

ATTIVO

Amministratore Admin Esci

**Attenzione** — funzione che consuma banda in continuo. La raccolta apre una sottoscrizione AISStream su **tutto il globo** (~100–300 messaggi/s, stimati ~200–400 MB/ora) e gira in background finché un admin non la ferma. Vengono salvate **solo** le densità per cella (nessun nome nave, nessuna posizione). Richiede la chiave di un **account AISStream separato**.

STATO

Ferma

BANDA (ORA)

0 B/s

SCARICATO

0 B

MESSAGGI/S

0

MESSAGGI SESSIONE

0

CONNESSIONE

—

CELLE POPOLATE

47.371

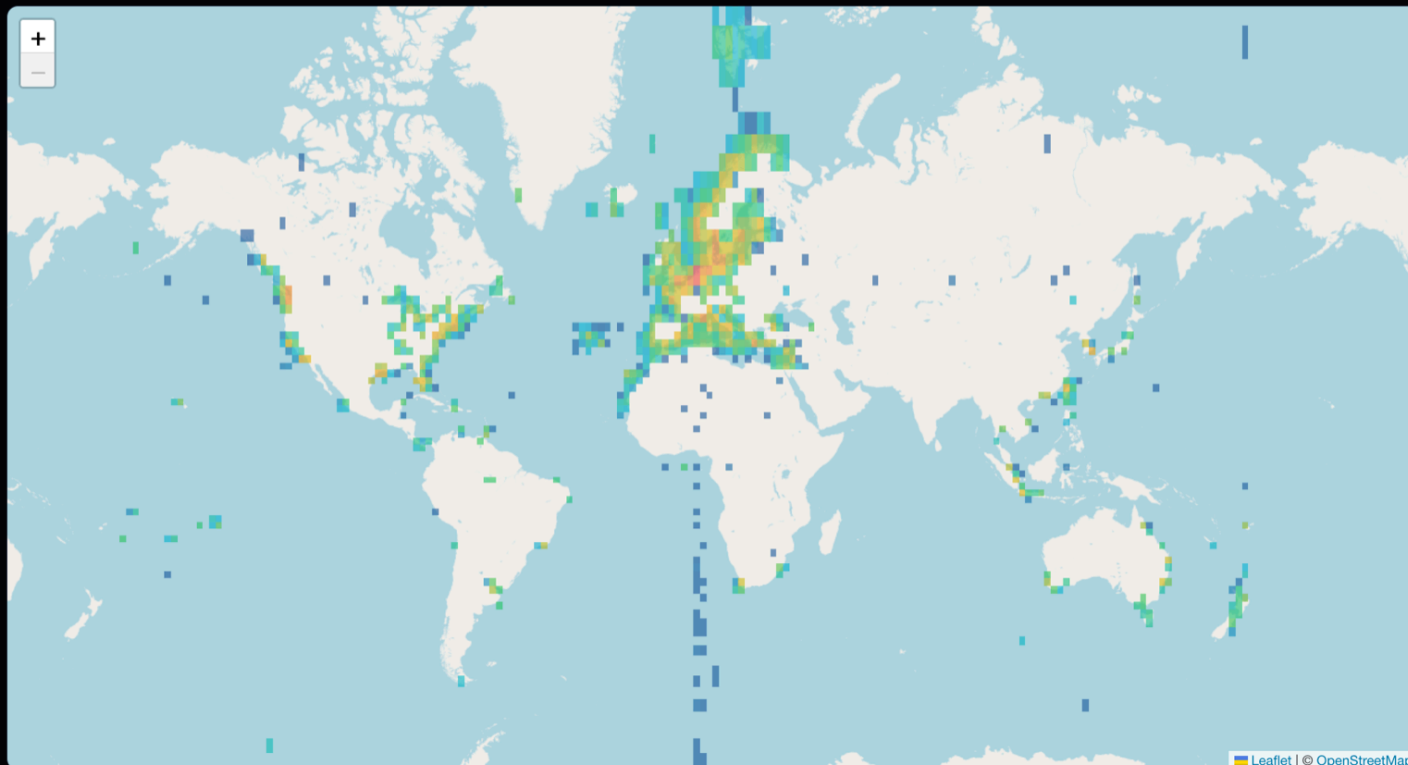
MESSAGGI TOTALI

1.969.821

▶ Avvia raccolta

↻ Aggiorna mappa

🗑 Cancelli dati



World coverage map: grid colored from blue (few AIS messages) to red (many).

Opens from  **Coverage map**. Shows a **world map** where each cell is colored according to **how many AIS messages are received** in that zone: from **blue** (few) to **red** (many). Useful for seeing at a glance where AIS coverage is good and where there are “holes”.

As a user, you can **open the map and see the current data** (read-only for you). The map is also available **without login** at `/heatmap`. Starting and stopping data collection is reserved to administrators.

A  button in the top-right corner (hover over it to see what it does) hides — **on by default** — cells with a **single message**: almost always isolated noise (e.g. satellite positioning artifacts far from any real route) rather than real coverage. Your choice is remembered for future visits.



# Settings

Open with ⚙ **Settings**. Settings are organized into **tabs**: **General**, **Areas**, **External integrations**, **Parameters**, **Backup / Restore**, and the technical 📡 **AIS Diagnostics** tab.

General tab

Tracking porti - Lido di venezia

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Import VesselFinder

Recupera informazioni aggiuntive da VesselFinder (bandiera, GT, DWT, anno costruzione, ecc.). Dati in cache per 6 ore.

Import MarineTraffic

Recupera informazioni aggiuntive da MarineTraffic nel dettaglio nave. Dati in cache per 6 ore.

Import ShipFinder

Recupera dati da ShipFinder e — unico tra le fonti — la posizione dell'ultimo avvistamento, usata per ri-localizzare le navi seguite che l'AIS non vede più. Dati statici in cache per 6 ore.

Import MyShipTracking

Come ShipFinder, recupera la posizione dell'ultimo avvistamento (più dati statici), usata per ri-localizzare le navi seguite che l'AIS non vede più: una seconda fonte di backup indipendente. Dati statici in cache per 6 ore.

Parametri scraping SF / MST

Intervallo minimo tra due scraping consecutivi per la stessa nave (minuti) e soglia di clustering posizioni (metri): se la nuova posizione è entro questa distanza dall'ultima memorizzata, aggiorna solo il timestamp invece di creare un nuovo punto. Zero = clustering disabilitato.

SF intervallo (min)

MST intervallo (min)

Cluster radius (m)

Lookup Equasis (proprietà)

Abilita il pulsante "Recupera informazioni Equasis" nel dettaglio nave per recuperare proprietario registrato, gestore ISM e operatore (per numero IMO). Mai automatico: parte solo su richiesta. Richiede credenziali Equasis (EQUASIS\_USER / EQUASIS\_PASSWORD in local.properties). Dati memorizzati una sola volta.

Visualizza log Equasis

Global Fishing Watch

Arricchisce le navi con identità ed eventi comportamentali di Global Fishing Watch (incontri in mare, loitering, scali in porto, eventi AIS spento), derivati dal feed AIS globale. Gli eventi alimentano il punteggio di rischio come conferme autoritative dei segnali comportamentali. Automatico come VesselFinder/MarineTraffic. Richiede un token API GFW (GLOBAL\_FISHING\_WATCH\_TOKEN in local.properties). Uso gratuito solo non commerciale.

Screening sanzioni

Confronta ogni nave con la lista sanzioni OFAC SDN (match per IMO, nome o call sign). Un match è un segnale di rischio molto forte. Lista sanzionata e aggiornata ogni 24h.

Settings — General tab: data source toggles, sanctions/PSC screening, notifications, and map overlay.

At the top, the **Area monitoring** panel shows all areas with a toggle to start/stop each one's stream (● active / ● off): this way you monitor several areas together.

Below, the toggles for **data sources and features**:

| Option                                                                                          | Function                                                                                                                          |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>VesselFinder / MarineTraffic</b>                                                             | Fetches additional data in the ship detail. Cache 6 hours.                                                                        |
| <b>Import ShipFinder</b>                                                                        | Data + <b>last position</b> to re-locate lost followed ships (orange markers). Off by default.                                    |
| <b>Import MyShipTracking</b>                                                                    | Second, independent position source (teal markers). Off by default.                                                               |
| <b>Sanctions screening</b>                                                                      | Matches every ship against the OFAC SDN list (by IMO/name/call sign). Refreshed every 24h; <b>Refresh list</b> forces a download. |
| <b>Additional sanctions lists (EU / UK / UN)</b>                                                | Adds the EU, UK OFSI, and UN lists (via OpenSanctions). Only active with sanctions screening on. Default on.                      |
| <b>Port State Control screening (Paris/Tokyo MoU)</b>                                           | White/grey/black flag performance + Paris MoU banned ships.                                                                       |
| <b>Equasis lookup (ownership)</b>                                                               | Enables the on-demand button in the detail. Never automatic.                                                                      |
| <b>Global Fishing Watch</b>                                                                     | Identity + behavioral events (proactive). On by default.                                                                          |
| <b>Notifications</b>                                                                            | General on/off switch for in-app notifications.                                                                                   |
| <b>Ship return / area change / high score / new berth / berth characterization notification</b> | Enables each individual in-app notification category.                                                                             |
| <b>Exclude tankers</b>                                                                          | Doesn't assign the "ship type" score to tanker hulls (useful when monitoring weapons transport).                                  |
| <b>Check position jump / Check AIS blackout</b>                                                 | Includes these signals in the risk score. Disable them in areas with poor AIS coverage (false positives). Default on.             |
| <b>OpenSeaMap nautical layer (tile)</b>                                                         | Nautical symbols as a single image (all or nothing). Default on.                                                                  |

| Option                                          | Function                                                                                                                                                                                             |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OpenSeaMap markers (selectable) + Items to show |  markers filterable by category (ports, moorings, anchorages, marinas, lighthouses, buoys, hazards...). Default on. |
| Risk weights by cargo type                      | Points assigned to each cargo class. Edit and  <b>Save weights</b> (immediate effect).                            |
| ↓ Export CSV                                    | Downloads all readings as CSV.                                                                                                                                                                       |
| ↓ Download backup / ↑ Restore                   | Downloads/reloads the database file.                                                                                                                                                                 |
| Language                                        | Italiano / English.                                                                                                                                                                                  |

**Warning: restoring** the database replaces **all** current data and is irreversible. Download a backup before proceeding. After restoring, data is reassigned to the correct area based on coordinates.

## Areas tab

Tracking porti - Lido di venezia

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Diagnostica AIS

Monitoraggio aree

Accendi o spegni lo stream AIS di ciascuna area. Più aree possono essere monitorate in contemporanea.

● Porto di Bari

● Israele

● Lido di venezia

● Nord Adriatico

● Porto di Taranto

Settings — Areas tab: list of areas with monitoring toggles.

Shows the configured areas with monitoring toggles (equivalent to the panel at the top of General) and links to area management.

## External integrations tab (Telegram + webhooks)

Tracking porti - Lido di venezia

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Notifiche Telegram

Ricevi le notifiche del tuo utente su Telegram tramite il bot. Collega il tuo account, poi scegli quali notifiche ricevere. Sono indipendenti dalle notifiche nella barra laterale.

Collegamento account

Account collegato. Ricevi le notifiche su Telegram.

Collega

Invia prova

Scollega

Telegram · score alto

Invia su Telegram l'arrivo di una nave con score di rischio in fascia rossa (71–100).

Telegram · rientro nave

Invia su Telegram il rientro di una nave in un'area già visitata.

Telegram · cambio area

Invia su Telegram lo spostamento di una nave tra due aree monitorate.

Telegram · nuova banchina

Invia su Telegram il rilevamento di una nuova banchina in un'area.

Telegram · caratterizzazione banchina

Invia su Telegram la prima caratterizzazione di una banchina.

Telegram · rendezvous in mare

Invia su Telegram il rilevamento di un rendezvous nave-nave (con mappa a due punti).

Telegram · disservizio AIS

Invia su Telegram l'inizio e la fine di un disservizio del feed AIS.

Telegram · avvio/stop monitoraggio area

Invia su Telegram quando avvii o fermi il monitoraggio di una tua area.

Telegram · mappa del punto

Allega alle notifiche con posizione (banchine e navi) un'immagine della mappa e un segnaposto toccabile.

Settings — External integrations tab: Telegram link and outgoing webhook configuration.

Here you connect the external channels notifications are sent to: **Telegram** (top) and **outgoing webhooks** (bottom).

**Linking Telegram** (works if the administrator has configured the bot):

1. Press **Link**. A link (and a code) appears.
2. Open the link on Telegram (or send the bot `/start <code>` ) and start the bot.
3. The bot replies “Account linked” and the tab refreshes: you now receive notifications on Telegram.

To stop, press **Unlink** (or `/stop` ). Use **Send test** to check the link.

**Which notifications to receive** — the **Telegram notifications** switch turns everything on/off; below it, a toggle per category: **High score**, **Ship return**, **Area change**, **New berth**, **Berth characterization**, **AIS outage**, **Area monitoring start/stop**. The **Position map** toggle attaches a **map image** plus a tappable pin to notifications that include a position. Telegram toggles are **independent** from the in-app ones.

Ship notifications (High score, Return, Area change) show 🗉 **Follow** and ★ **Flag** buttons on Telegram to act directly from the message.

🔗 **Outgoing webhooks** — forwards events from your areas to a web address (Slack, Discord, a SIEM, or your own service):

1. Paste the webhook **URL**.
2. Choose the **format**: *Generic* (raw JSON), *Slack*, or *Discord*.
3. Check **which events** to send (high risk, rendezvous, area change, return, berths, AIS outage).
4. (Optional) set a **secret**: adds an `X-Tracker-Signature` header the recipient can verify.
5. **Add webhook**. **Test** sends a test event; the switch enables/disables it; **Delete** removes it.

Webhooks are **personal** (only for your areas). Internal/private addresses aren't allowed. Maximum 10 per user.

# Parameters tab

Tracking porti - Lido di venezia

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⚠

Le modifiche a questi parametri vengono scritte in `app.config.properties` e richiedono il **riavvio del server** per avere effetto (non basta ricaricare il browser). I segreti e le credenziali restano nel file `local.properties` ; i toggle di import e notifiche sono nel tab **Generali**.

📄

Salva parametri

2 modifiche non salvate

RILEVAMENTO STATO NAVE

Soglia SOG (nodi) sotto cui la nave è considerata ferma

SOG\_FERMA\_KN

0,5

kn

Raggio (metri): ultime posizioni entro questo raggio → nave ormeggiata (isteresi anti-swing)

STILL\_RADIUS\_M

100

m

FINESTRE DI VISIBILITÀ

Ore: una nave in movimento viene mostrata in "Navi presenti" se vista in questa finestra

ACTIVE\_WINDOW\_HOURS

6

ore

Ore: una nave in porto (ormeggiata/all'ancora/ferma) resta in "Navi presenti" più a lungo

PORT\_WINDOW\_HOURS

24

ore

CONNESSIONE E POLLING

Millisecondi: attesa prima di riconnettersi ad AISStream dopo un errore

RECONNECT\_DELAY\_MS

5000

ms

Millisecondi: intervallo di aggiornamento del frontend (default 5 minuti)

POLL\_INTERVAL\_MS

300000

ms

Rilevamento disservizio AIS: se uno stream attivo non riceve messaggi per AIS\_OUTAGE\_SILENCE\_MIN minuti, l'app interroga un monitor di uptime AISStream (progetto MIT [github.com/buttermilkgreen/AISStream-Uptime](https://github.com/buttermilkgreen/AISStream-Uptime)) tramite la sua API /api/v1/status. Solo se anche quel monitor riporta il servizio non attivo viene

Settings — Parameters tab: configuration fields grouped by category with description.

Lets you edit the app’s **operating parameters** (ship status thresholds, time windows, retention, berths, score weights...) from the interface. Every field has a description. Edit the values and press  **Save parameters**.

 **Important:** these parameters are read by the server **only at startup**. After saving, you **must restart the server** for the changes to take effect — reloading the browser is not enough. **Secrets** (API keys, passwords) can't be edited here for security.



# Backup / Restore tab

Tracking porti - Lido di venezia

• ATTIVO

← Indietro

GeneraliAreeIntegrazioni esterneParametriBackup / RipristinoLog attivitàLog APIDiagnostica AIS

AUTO-BACKUP LOCALE

Backup automatico

Backup automatico ogni 2 ore, salvato sul server. Vengono conservati gli ultimi 5. Puoi anche salvare un backup manuale ora.

Salva ora

Auto

16/07/2026, 12:38:07 · 298 KB

Ripristina

BACKUP COMPLETO

Esporta tutto

Scarica un unico file con database, aree e impostazioni. Re-importabile per ripristinare l'intero stato dell'app.

Esporta tutto

Importa tutto

Ripristina database, aree e impostazioni da un file di backup completo. Sostituisce **tutti** i dati attuali. Operazione irreversibile.

Importa tutto

DATI

Esporta CSV

Scarica tutte le letture AIS in formato CSV.

Esporta

Backup database

Scarica l'intero database (file .db) come backup locale.

Scarica

Ripristina database

Sostituisce **tutti** i dati attuali con quelli del file di backup selezionato. Operazione irreversibile.

Ripristina

MAPPA ZONE COPERTE

Esporta dati copertura

Scarica il database separato della mappa delle zone coperte (file .db). È incluso anche nel backup completo.

Esporta

Importa dati copertura

Sostituisce **tutti** i dati attuali della mappa delle zone coperte con quelli del file selezionato.

Importa

Settings — Backup tab: download and export data, restore a backup.

Here you **download a backup** of the database, **restore** a saved backup, and **export** data. The Coverage map data lives in a separate database, exportable/importable on its own and still included in the full backup.

# AIS Diagnostics tab

Tracking porti - Lido di venezia

• ATTIVO

AmministratoreAdminEsci

← Indietro

GeneraliAreeIntegrazioni esterneParametriBackup / RipristinoLog attivitàLog APIDiagnostica AIS

Monitoraggio aree (lido\_di\_venezia)

Chiave: ...0950 (env) - env

CONNESSIONE

● Connesso 12:37:37 16/07/2026

FRAME WS RICEVUTI

170

VELOCITÀ MESSAGGI

18 msg/min

LETTURE TOTALI DB

33.942

UPTIME SESSIONE

9m 40s

MESSAGGI NAVE (SESSIONE)

170

TOTALE RICONNESSIONI

0

ULTIMO ERRORE AIS

Nessuno

Navi seguite

Chiave: ...3d34 (env) - env

CONNESSIONE

● Connesso 12:42:37 16/07/2026

FRAME WS RICEVUTI

5

VELOCITÀ MESSAGGI

1 msg/min

NAVI SEGUITE

7

ULTIMO ERRORE AIS

Nessuno

UPTIME SESSIONE

4m 40s

MESSAGGI NAVE (SESSIONE)

5

TOTALE RICONNESSIONI

0

IN RI-ACQUISIZIONE

6

Mappa zone coperte

Chiave: ...6d04 (env) - env

CONNESSIONE

● Disconnesso

FRAME WS RICEVUTI

0

VELOCITÀ MESSAGGI

UPTIME SESSIONE

0s

MESSAGGI NAVE (SESSIONE)

0

TOTALE RICONNESSIONI

Settings — AIS Diagnostics tab: connection status, uptime, frames received, and reconnections.

Shows the data stream connection status (refreshes every 5s):

- **Connection** — Connected / Disconnected
- **Session uptime** — how long the stream has been active
- **WS frames received / Ship messages / Message rate**
- **Reconnections** — how many times the connection was restored
- **Last error** — the most recent error, if any

### AIS outage banner

If an active monitoring session receives **no AIS signal at all** for a few minutes, the app checks the service status with an independent uptime monitor. Only if that also confirms the outage does a yellow warning appear at the top of monitoring pages. If the area is simply quiet but the service is up, no warning appears.

The same warning also appears if a monitoring session or the **followed ships** stream gets stuck repeatedly reconnecting for a few minutes — never stabilizing, even when each individual attempt only lasts a few seconds (in this case, no external confirmation is needed, since one of our own connections failing to stabilize is an unambiguous signal). In both cases you can dismiss the warning with **✕**; it disappears on its own once the connection stabilizes.

### Risk score

Every ship receives a score from 0 to 100, computed automatically. It is indicative and does not replace expert assessment.

| Color  | Band   | Meaning                 |
|--------|--------|-------------------------|
| Green  | 0–30   | Low risk                |
| Yellow | 31–70  | Medium risk — monitor   |
| Red    | 71–100 | High risk — investigate |

### Source indicators on the badge:

- **Magenta** dot: computed with VesselFinder data
- **Gold** dot: MarineTraffic data
- **Orange** dot: both sources
- **Red** dot (with halo): ship on a sanctions list (OFAC / EU / UK / UN)
- **Blue** label (Paris/Tokyo MoU ⚓ ): signal from Port State Control lists
- **Teal** dot: Global Fishing Watch data

Hover over the badge for factor and source details.

**Weight by cargo type:** one factor depends on the ship’s cargo class, with configurable weights in ⚙ **Settings** → **“Risk weights by cargo type”** (immediate effect). With **“Exclude tankers”** on, classes on tanker hulls don’t assign points.

**Military ships:** automatically at maximum risk (red row).

## Notifications

Besides temporary on-screen alerts, the app keeps a **notification history** in the sidebar, opened/closed with  **Notifications** (the state is remembered).

**When a notification is generated** (each category can be enabled separately from [Settings](#)):

Ship events:

- **Ship return** — a ship already seen in an area is detected again in the same area.
- **Area change** — a ship seen in one area is detected in a **different** one.
- **High score** — a ship arrives with a score in the red band (71–100).

Berth events:

- **New berth** — a new berth is detected during recomputation.
- **Berth characterization** — a berth is characterized for the first time.

Other events:

- **Rendezvous at sea** — two ships stay close, slow, and offshore long enough (possible transshipment). The notification includes a map with the two points joined by a line.

**Reading a notification:**

| Element                                                                                                                                                                                                                                                             | Meaning                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Dot  /  /  | Color for risk band (ship notifications); berth notifications have a dedicated dot   |
| Text                                                                                                                                                                                                                                                                | Ship name and area, or origin/destination area (area change), or berth name/category |
| ✓ button                                                                                                                                                                                                                                                            | Mark as read                                                                         |
|  button                                                                                                                                                                           | Delete (warning with ↶ <b>Undo</b> for 5s)                                           |

**Click the notification** (outside the buttons): for a ship, opens its card; for a berth, switches to the area map and centers on the berth.

**Muting a single ship:** in the detail, the 🔔 (active → click to mute) / 🔔 (muted → click to re-enable) button. A muted ship generates no return or area-change notifications.

**Unread** notifications are bold and count toward the red badge. The **last 100** are kept; older ones are deleted. Clearing an area's data also removes its notifications.

## Exporting data

All exports are downloaded directly from the browser:

- **CSV** — **↓ Filtered CSV** in the Current/Past ships toolbar exports the current view (filtered and sorted); ⚙ **Settings** → **Export CSV** instead exports all raw readings.
- **GeoJSON / KML** (for **QGIS** or **Google Earth**) — next to the CSV button you'll find **↓ GeoJSON** and **↓ KML**. Four sources: the filtered **ship list** (points), a ship's **track** (from the detail), an area's **replay** (one line per ship), and **berths** (polygons).

## Installing the app (PWA)

Tracker Porti is an **installable app** (PWA): you can add it to your phone's home screen or install it on desktop, and it opens full-screen.

- **On phone** — browser → menu → **"Add to Home Screen"** (iPhone/Safari) or **"Install app"** (Android/Chrome).
- **On desktop** — in Chrome/Edge, the install icon in the address bar, or menu → **"Install Tracker Porti"**.

If there's no connection, the app shows an **"You're offline"** screen with a *Retry* button (AIS data is real-time and needs the network). Access stays protected: login is always required.

## Frequently asked questions

**The table is empty — what do I do?** Check that monitoring for the area is started (● **ACTIVE** badge at the top) and that the area has ship traffic. Check the monitored areas from the "Area monitoring" panel in Settings,

and the connection from **AIS Diagnostics**.

**How do I avoid losing track of ships I've already checked?** Use ✓ **Seen** on each row: the ship dims, distinguishing it from ones not yet reviewed.

**I changed area and the ships disappeared — is that normal?** Yes. Each area has independent data and stream. Changing area in the menu is just a view switch: it doesn't start or stop any stream. The previous area's data stays; go back to see it again.

**Can I monitor several areas at once?** Yes. ⚙ **Settings** → **Area monitoring** and turn on each area's toggle. Then switch between them with the dropdown menu.

**Are military ships always red?** Yes. They're automatically flagged with maximum score and a red row.

**Can I export the data?** Yes: CSV, GeoJSON, and KML — see [Exporting data](#).